

LLM Knowledge Distillation

(Teacher → Student)

White-Box Distillation

Full access to model internals
(weights, activations, attention, hidden states)

Layer-wise Distillation



Match hidden states & embeddings

Attention Transfer



Copy attention matrices & patterns

Feature Matching



Intermediate representations

Examples: TinyBERT, DistilBERT, MiniLM

Black-Box Distillation

API access only
(only text inputs/outputs visible)

Instruction Following

Q: "Explain AI"

Teacher response

Train student on
(Q, A) pairs

Chain-of-Thought Distillation

Step 1 →

Step 2 →

Answer

Copy reasoning process

In-Context Learning

Example 1

Example 2

Learn from demonstrations

Examples: Self-Instruct, Alpaca, Vicuna

Key Differences

White-Box

- Full model transparency
- Rich gradient information
- Better performance potential
- Requires open-source model
- Higher computational cost

Black-Box

- Only output text visible
- Works with proprietary APIs
- More practical for GPT-4, etc.
 - Simpler implementation
- Less precise knowledge transfer